

SIP call-flow cheat sheet

Registration

1. REGISTER to the domain (Request-URI), To = AOR, Contact = device.
2. 401 and WWW-Authenticate. A proxy would send 407 instead.
3. REGISTER with Authorization. New CSeq.
4. 200 OK stores the binding. Expires: 0 removes one. Contact: * and Expires: 0 removes all.

Basic call

1. INVITE (early offer puts SDP here).
2. 100 Trying is hop-by-hop.
3. 180 Ringing is alerting. A To tag means an early dialog, not answer.
4. 200 OK confirms the dialog. ACK for that 2xx is a new end-to-end transaction.
5. BYE ends it. Record-Route keeps the proxy on ACK and BYE.

ACK for a non-2xx final response stays inside the INVITE transaction.

UC Lab field card · fictional examples only · confirm RFCs and current vendor docs before production changes.

Hold (RFC 3264)

re-INVITE, same Call-ID, higher CSeq. sendonly is answered reconly. inactive is answered inactive. Port 0 rejects the stream; it is not hold. Avoid c=0.0.0.0.

Transfer

REFER, then 202. 202 is not success. NOTIFY carries message/sipfrag status. Blind: Refer-To is a URI. Attended: Refer-To includes Replaces (Call-ID and tags). RFC 3515, RFC 3891, RFC 5589.

Early media and PRACK

180 with no SDP: local ringback. 183 with SDP: play the RTP. PRACK acknowledges a reliable 1xx. RACK echoes RSeq plus the INVITE CSeq. The 200 to PRACK is not the 200 to INVITE.

Refresh

re-INVITE or UPDATE. Session-Expires and refresher come from RFC 4028. A rejected re-INVITE leaves the last good session in place. 491 means glare.